Token-World: Direct World Model Simulator for Vision-Language-Action Models

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Abstract

World-model simulators for vision-language-action (VLA) systems often predict future RGB frames and then re-encode them into policy inputs. This indirect decode–encode interface creates a representation mismatch between the simulator and the downstream policy, adding computational overhead and potential token-level information loss to every imagined interaction.To address this bottleneck, we introduce **Token-World**, an action-conditioned world model that operates directly in VLA visual token space. Token-World uses a spatio-temporal multimodal Transformer with an x_0-parameterized flow-matching objective to model high-dimensional semantic token dynamics, together with a GRU-based recurrent state that preserves past interaction context during long-horizon rollout. Experiments evaluate Token-World along two complementary axes: for token-dynamics modeling, the 2D flow-matching Transformer improves token prediction accuracy over deterministic Transformer-based baselines; for the simulation interface, direct token-space rollout outperforms RGB-based world models in long-horizon prediction accuracy, inference efficiency, and rollout-based success classification.

1 Introduction

World models [18] offer a promising route toward scalable embodied learning by serving as learned simulators of action-conditioned environment dynamics [14, 19, 20, 51, 53]. Instead of executing every behavior in the physical world, an agent can roll out future observations inside the model, which is especially valuable in robotics where real-world interaction is costly and slow.[13, 51]. Recent work has therefore explored world models as simulators from several perspectives: using predicted futures to evaluate policies[16, 33, 34, 38, 47], synthesizing additional interaction data for policy learning[17, 27, 45, 49], and performing reinforcement learning through imagined rollouts[8, 28, 29, 39, 52, 57].

As world models are increasingly used as learned simulators, they are often paired with VLA policies that map visual observations and language instructions to actions [7, 26, 31, 46]. However, existing pipelines often use an indirect simulation interface, where future RGB observations are first predicted by the world model, then re-encoded into VLA visual tokens before being consumed by the VLA policy [8, 16, 17, 27–29, 33, 34, 38, 39, 45, 47, 49, 52, 57]. As world models move toward scalable simulators for VLA agents, this indirect interface becomes a critical bottleneck: the simulator predicts human-viewable pixels, whereas the downstream policy ultimately consumes policy-facing visual tokens.This mismatch is not merely computational: RGB reconstruction encourages the simulator to allocate capacity to visual details that may be irrelevant to the policy, while token-space rollout directly targets the representation used for downstream action generation.

A representation-aligned alternative is to simulate the future directly in the VLA visual-token space consumed by the policy. This direct interface bypasses the decode–encode pipeline, avoiding repeated

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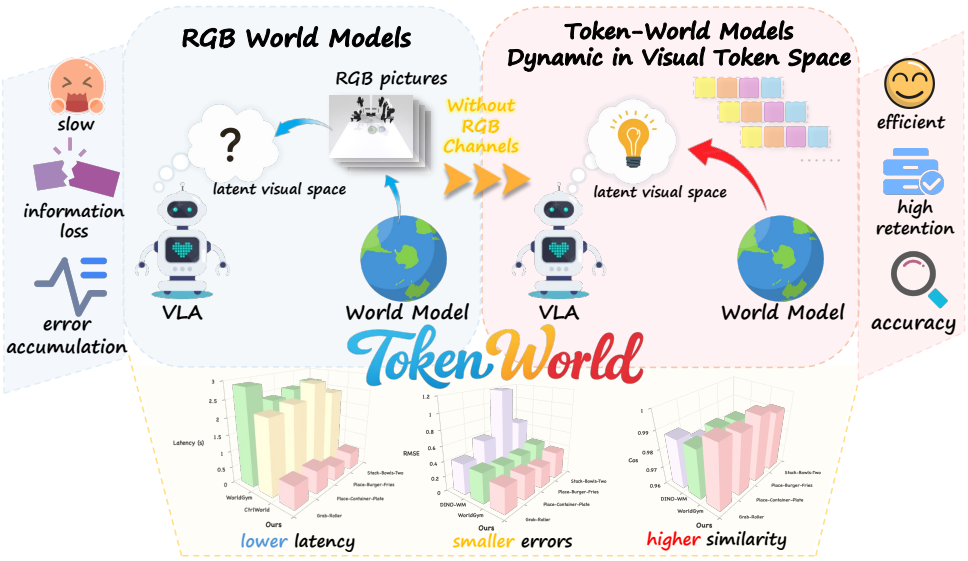


Figure 1: **Token-World simulates directly in VLA visual-token space.** RGB-based simulators predict future frames and re-encode them into VLA visual tokens, introducing a decode–encode interface. Token-World rolls out directly in the policy-facing token space, yielding more faithful token predictions with about $4\times$ lower simulator step time.

reconstruction and re-tokenization during imagined rollouts. However, direct VLA visual-token simulation is non-trivial: these tokens are compact, policy-facing representations [7, 26, 31], and predicting their future evolution requires modeling high-dimensional dynamics while retaining past interaction context [50]. Therefore, both the dynamics architecture and the learning objective must be designed around the specific challenges of token-space rollout.

In this work, we introduce Token-World, a world model simulator that directly operates in VLA visual token space. To model high-dimensional VLA visual token dynamics, we use a spatio-temporal multimodal Transformer [21, 23, 48] that efficiently captures within-frame token interactions and cross-frame temporal evolution. Following [21, 32], Token-World uses x_0-parameterized flow-matching objective to better model the high-dimensional token dynamics. Moreover, we introduce a GRU-based recurrent state [11] to preserve the history-dependent context of VLA visual tokens[19, 20].

Our experiments disentangle two key design choices. First, for token-dynamics modeling, the 2D flow-matching Transformer improves token prediction accuracy over deterministic Transformer-based baselines. Second, for the simulation interface, direct token-space rollout achieves better long-horizon prediction accuracy, higher inference efficiency, and more reliable rollout-based success classification than RGB-based world models. In summary, our contributions are threefold:

- We identify the decode–encode interface as a representation mismatch in world-model simulators for VLA systems, formalize this interface problem in Sec. 3, and propose direct VLA visual-token simulation as a policy-aligned world-model interface.
- We present Token-World, a 2D flow-matching Transformer world model simulator with x_0-parameterized flow-matching objective to model high-dimensional semantic token dynamics, together with a GRU-based recurrent state to retain past interaction context during long-horizon rollout.
- We provide a two-axis empirical validation of Token-World. On the modeling axis, our 2D flow-matching Transformer formulation achieves more accurate VLA visual-token prediction than deterministic Transformer baselines. On the interface axis, direct token-space rollout improves long-horizon prediction accuracy, inference efficiency, and rollout-based success classification over RGB-based world-model simulators.

2 Related Works

2.1 World Models as Learned Simulators

World models learn action-conditioned dynamics and support planning or policy learning through imagined rollouts [14, 18–20, 51, 53]. In robotics, recent work has developed learned simulators for several purposes: evaluating candidate actions or policies with predicted futures [16, 33, 34, 38, 47], generating synthetic interaction data for policy learning [17, 27, 45, 49], and performing reinforcement learning inside learned dynamics models [8, 28, 29, 39, 52, 57]. These methods reduce the need for costly physical interaction and point toward learned world models as a promising path to scalable simulation for embodied agents.

Most of these simulators, however, represent imagined futures as RGB frames. Token-World is complementary to this line of work: instead of proposing a new downstream use of imagined rollouts, it studies the simulation interface itself, directly rolling out future states in the VLA visual-token space consumed by downstream policies.

2.2 Visual Representations for World Models

A key design choice in visual world models is the representation space used for future prediction. A prominent class of recent embodied world models adopts a latent video-generation-style formulation, either by adapting pretrained video generation models or by training similar architectures from scratch. These methods model future dynamics in compressed VAE or video-token latent spaces and decode them back into RGB/RGB-D frames for visual rollout [8, 16, 17, 22, 24, 27–29, 33–35, 38, 45, 47, 49, 52, 55, 57]. Another line of work avoids reconstructing redundant RGB observations and instead models dynamics in representation space. DINO-based methods predict the evolution of pretrained visual features for future scene understanding, planning, or robotic manipulation [3, 30, 56]. In parallel, JEPA-style world models formulate future prediction as latent embedding prediction, enabling efficient planning in learned representation spaces [2, 4, 36, 37, 41, 43]. While both directions move beyond RGB reconstruction, their prediction spaces are usually generic visual representations rather than the policy-facing visual-token space consumed by VLA systems.

Token-World differs from previous work in both representation and dynamics modeling. In representation, we target VLA visual tokens, a policy-facing space directly consumed by downstream VLA agents. In dynamics modeling, we use a 2D flow-matching Transformer architecture with a recurrent state, instead of the deterministic Transformer predictors commonly used in DINO-WM [56], V-JEPA2 [2], and related latent world models.

3 Problem Formulation

Information loss induced by decode–encode interfaces. Many world-model simulators operate in a reconstruction-oriented latent space and interface with a VLA policy through a decode–encode pipeline. Let o_t denote the original observation and let $z_t = E(o_t)$ be the policy-facing VLA visual tokens produced by a frozen VLA visual encoder E. An RGB/VAE-based world model first encodes the observation into a latent variable

$$u_t \sim q_\psi(u_t \mid o_t), \tag{1}$$

and reconstructs an image through a decoder

$$\hat{o}_t = D_\theta(u_t). \tag{2}$$

The VLA policy cannot directly consume u_t or $\hat{o}_t$; instead, the reconstructed image must be re-encoded into visual tokens:

$$\hat{z}_t = E(\hat{o}_t) = E(D_\theta(u_t)). \tag{3}$$

This induces the following data-processing chain:

$$o_t \to u_t \to \hat{o}_t \to \hat{z}_t, \tag{4}$$

while the desired policy-facing token is $z_t = E(o_t)$.

The VAE latent u_t is explicitly rate-limited by its variational objective. For a β-VAE-style formulation,

$$\mathcal{L}_{\mathrm{VAE}} = \mathbb{E}_{q_\psi(u_t|o_t)}\big[-\log p_\theta(o_t \mid u_t)\big] + \beta\,\mathrm{KL}\big(q_\psi(u_t \mid o_t)\,\|\,p(u_t)\big), \tag{5}$$

the KL term upper-bounds the mutual information between the observation and the latent variable:

$$I(o_t; u_t) \le \mathbb{E}_{p(o_t)} \mathrm{KL}\big(q_\psi(u_t \mid o_t) \,\|\, p(u_t)\big). \tag{6}$$

Thus, under a finite-rate latent bottleneck, u_t cannot preserve all information in o_t. Since $\hat{o}_t$ is generated only from u_t, the data processing inequality gives

$$I(o_t; \hat{o}_t) \le I(o_t; u_t), \tag{7}$$

which formalizes the information loss introduced by reconstructing the image from a compressed latent representation.

More importantly for VLA policies, the decode–encode interface can only preserve token information that survives this reconstruction process. Since $z_t = E(o_t)$ and $\hat{z}_t = E(\hat{o}_t)$, we have

$$I(z_t; \hat{z}_t) \le I(z_t; \hat{o}_t) \le I(o_t; \hat{o}_t) \le I(o_t; u_t). \tag{8}$$

Combining the above inequalities, we obtain a direct upper bound on the token information preserved by the decode–encode interface:

$$I(z_t; \hat{z}_t) \le \mathbb{E}_{p(o_t)} \mathrm{KL}\big(q_\psi(u_t \mid o_t) \,\|\, p(u_t)\big). \tag{9}$$

This bound makes the interface bottleneck explicit: the mutual information between the true policy-facing token z_t and the re-encoded token $\hat{z}_t$ cannot exceed the information rate allowed by the VAE latent.

Thus, unless the VAE latent has sufficient capacity to preserve all information needed to recover the VLA token, the decode–encode interface inevitably leaves residual uncertainty about the policy-facing representation. During multi-step imagined rollout, the same bottleneck is repeatedly applied, allowing token-level errors to accumulate over time. This motivates bypassing the reconstruction interface altogether and directly modeling future states in the VLA visual token space.

Direct VLA visual-token simulation. To avoid the intermediate reconstruction bottleneck, Token-World directly models future dynamics in the VLA visual-token space. At each timestep, we treat the VLA visual tokens and proprioceptive state as the world-model state,

$$s_t = (z_t, p_t), \tag{10}$$

where z_t denotes the policy-facing visual tokens and p_t denotes the proprioceptive state. Given the past state history $s_{\le t}$ and the current action a_t, Token-World parameterizes a one-step state dynamics model:

$$p_\theta\big(z_{t+1}, p_{t+1} \mid z_{\le t},\, p_{\le t},\, a_t\big)\,. \tag{11}$$

Long-horizon rollout is then obtained autoregressively by repeatedly applying the one-step model under a future action sequence $a_{t:t+H-1}$:

$$p_\theta(z_{t+1:t+H}, p_{t+1:t+H} \mid z_{\le t},\, p_{\le t},\, a_{t:t+H-1}) = \prod_{k=0}^{H-1} p_\theta(z_{t+k+1}, p_{t+k+1} \mid \hat{z}_{\le t+k},\, \hat{p}_{\le t+k},\, a_{t+k})\,, \tag{12}$$

where $\hat{z}_{\le t+k}$ and $\hat{p}_{\le t+k}$ denote the rollout history formed by appending previously predicted visual tokens and proprioceptive states. The predicted visual tokens remain in the same visual-token space as the downstream VLA policy. Unlike RGB-based simulators, this direct interface avoids decoding latent states into RGB observations and re-encoding them with the VLA visual tokenizer.

4 Token-World

4.1 Overview

Token-World is an action-conditioned world model that directly operates in the VLA visual-token space. Following [21], we implement Token-World as a spatio-temporal multimodal Transformer [48], where each block factorizes modeling into within-frame spatial attention and causal temporal attention [23].To retain interaction context beyond the finite attention window, we further insert a GRU-based temporal state [11] along the temporal axis.

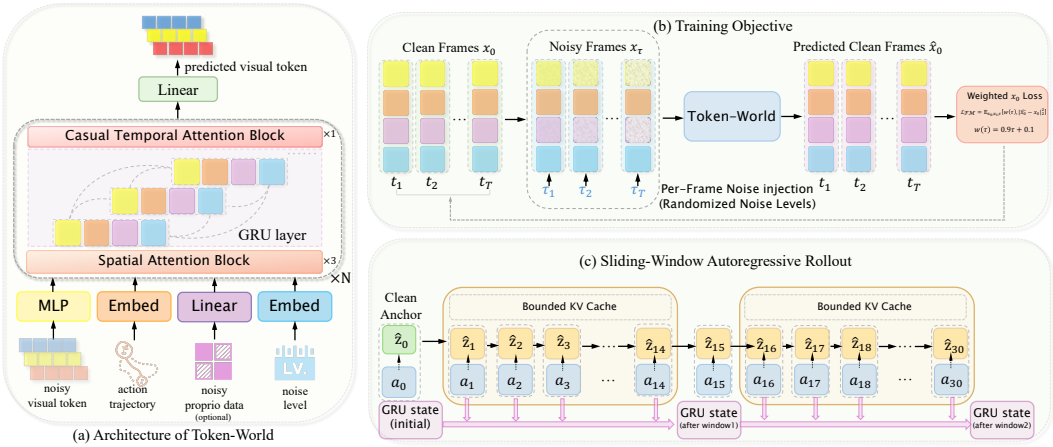


Figure 2: **Token-World overview. (a)** Architecture: a spatio-temporal multimodal Transformer models VLA visual-token dynamics, with a GRU state carrying context beyond the finite attention window. **(b)** Training: x_0 flow matching predicts clean future tokens from frame-level noisy inputs. **(c)** Inference: sliding-window autoregressive rollout generates future tokens directly in the VLA visual-token space.

We adopt an x_0-parameterized flow-matching objective to better model high-dimensional VLA visual token dynamics[32]. In implementation, we train the model through a shortcut forcing objective [9, 15, 21] to improve robustness under autoregressive generation and decrease inference time.

At inference time, Token-World performs windowed autoregressive rollout [9, 24, 25, 49]. Starting from a clean anchor token, the model predicts future VLA visual tokens step by step within a finite context window; when the window reaches its maximum length, the last denoised prediction is used as the anchor for the next window. This allows Token-World to produce long-horizon rollouts entirely in the VLA visual-token space while keeping the temporal context and memory cost bounded.

4.2 Architecture of Token-World

Spatio-Temporal Multimodal Transformer. Following [21], we organize the input as a tensor of shape (B, T, S, D), where T is the rollout-window length, S is the number of multimodal tokens per timestep, and D is the hidden dimension. At each timestep, VLA visual tokens are first projected into the model dimension, while proprioceptive states, actions, and noise level embeddings are mapped into the same token space. These tokens are concatenated to form a multimodal token set for each frame, as shown in Figure 2 (a).

Each Transformer block follows a causal spatio-temporal design over space and time. Tokens within the same frame can attend to each other, while temporal attention is causally masked so that each frame only attends to past context. To reduce the cost of dense attention over all space-time tokens, we decompose attention into space-only and time-only layers [23], and apply temporal attention sparsely across the network [21]. We use a standard pre-norm Transformer implementation with RMSNorm [54], RoPE [42], SwiGLU [40], QK normalization [12], attention logit soft-capping [5, 44], and grouped-query attention (GQA) [1] for stable and efficient training with KV caching.

GRU-based Recurrent State. The spatio-temporal Transformer captures dependencies within a finite attention window. During long-horizon rollout, this window is repeatedly shifted forward, causing earlier interaction context to fall outside the temporal attention range. We therefore introduce a GRU-based [11] recurrent state along the temporal axis. Unlike KV cache, which grows with the number of cached timesteps, the GRU state provides a fixed-size memory that summarizes past token-action context across windows.

Concretely, the hidden representation produced during temporal modeling is used to update a recurrent state that summarizes past visual-token, proprioceptive, and action context. This state is then fed back into the transition model as compact temporal memory. Unlike the temporal KV cache, which

stores token-level keys and values within the current attention window, the GRU state has fixed size and can carry interaction context across windows without growing with rollout length.

4.3 Training Objective and Autoregressive Rollout

Training Objective. Following [21, 32], we train the transition model with an x_0-parameterized flow-matching objective in the VLA visual token space, as shown in Figure 2 (b). Let $\mathbf{x}_0$ denote the clean target future world-model state, as defined in Sec. 3, and let $\mathbf{x}_1 \sim \mathcal{N}(0, \mathbf{I})$ denote Gaussian noise. For a flow time $\tau \in (0, 1]$, we construct

$$\mathbf{x}_\tau = (1 - \tau)\mathbf{x}_0 + \tau\mathbf{x}_1. \tag{13}$$

The transition model predicts the clean endpoint

$$\hat{\mathbf{x}}_0 = f_\theta(\mathbf{x}_\tau, \tau, \mathcal{C}), \tag{14}$$

where $\mathcal{C}$ denotes the context tokens, including visual-token history, proprioceptive states, actions, and the recurrent state. Following [21], we use a ramp weighting term to emphasize harder denoising steps with larger noise levels:

$$w(\tau) = 0.9\tau + 0.1. \tag{15}$$

The final training objective is

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}_{\mathbf{x}_0, \mathbf{x}_1, \tau}\left[w(\tau)\,\|\hat{\mathbf{x}}_0 - \mathbf{x}_0\|_2^2\right]. \tag{16}$$

In our windowed training setup, we adopt the shortcut forcing objective over rollout windows [9, 15, 21], assigning different noise levels to different rollout steps. This exposes the model to imperfect autoregressive histories during training and enables efficient inference with a reduced number of denoising steps.

Autoregressive Rollout. At inference time, Token-World performs windowed autoregressive rollout rather than denoising an entire future sequence in one pass [9, 24, 25, 49], as shown in Figure 2 (c). Each rollout window starts from a clean anchor token. For the first window, the anchor is the observed token z_t; for later windows, the anchor is the last denoised prediction from the previous window.

Within a window, future tokens are generated step by step. Let s_m be the starting timestep of the m-th window and L be the maximum window length. For the r-th prediction inside this window, Token-World samples

$$\hat{z}_{s_m+r+1} \sim p_\theta(\cdot \mid \tilde{z}_{s_m:s_m+r},\, p_{s_m:s_m+r},\, a_{s_m:s_m+r}), \quad r = 0, \ldots, L-2, \tag{17}$$

where $\tilde{z}_{s_m}$ is the clean anchor token and $\tilde{z}_{s_m+i} = \hat{z}_{s_m+i}$ for $i > 0$ are tokens generated earlier in the same window. After the window reaches its maximum length, the final predicted token is used as the clean anchor for the next window, and rollout continues with a shifted temporal context.

This procedure keeps the temporal attention window bounded while allowing long-horizon simulation. The GRU recurrent state is carried across windows, providing compact memory of earlier interaction context beyond the current attention range.

5 Experiments

5.1 Overview

Our experiments evaluate Token-World along two complementary axes. First, we study the dynamics modeling axis: given the VLA visual-token prediction target, what modeling choices are effective for VLA visual-token dynamics? Second, we study the simulation-interface axis: when the downstream policy consumes VLA visual tokens, does direct token-space simulation provide a better interface than RGB-based decode–encode simulation? We further include qualitative visualizations to analyze how predicted token changes align with ground-truth semantic dynamics over long-horizon rollouts.

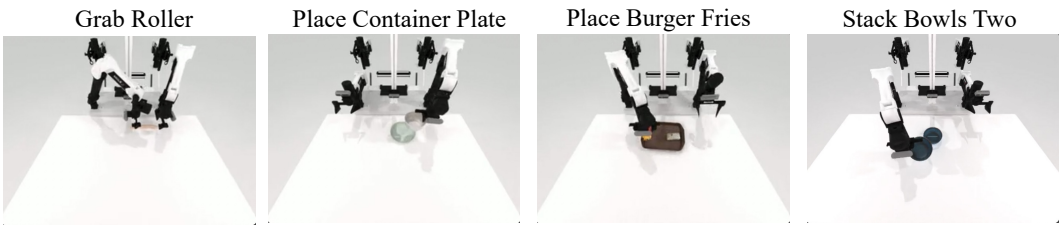


Figure 3: **The four manipulation tasks on RoboTwin benchmark.**

5.2 Experimental Setup

Tasks and data. We evaluate Token-World on four simulated manipulation tasks from RoboTwin [10]: *Grab-Roller*, *Place-Container-Plate*, *Place-Burger-Fries*, and *Stack-Bowls-Two*. For each task, we collect 100 trajectories with a balanced split of 50 successful and 50 failed episodes, except for *Grab-Roller*; details are provided in Appendix A. We split them into 80 training and 20 validation trajectories while preserving the success/failure balance. All world models are trained on the same task data under matched training budgets. We use the Paligemma [6] visual tokenizer from a task-specific π_0 [7] VLA policy as the frozen visual encoder.

Baselines. We group baselines by the two evaluation axes. For *dynamics modeling*, all methods predict VLA visual tokens directly; we compare Token-World with a deterministic Transformer token predictor adapted from DINO-WM [56]. For *simulation interface*, we compare with RGB-based simulators, WorldGym [38] and Ctrl-World [17]. WorldGym predicts VAE latents with a DiT dynamics model and decodes them to RGB, while Ctrl-World adapts a pretrained video diffusion backbone for controllable robot world modeling.

Metrics. We evaluate all models with open-loop action replay. Token fidelity is measured by RMSE and cosine similarity between predicted and ground-truth VLA visual tokens; efficiency is measured by simulator step time; rollout utility is measured by success prediction using a shared latent success classifier, detailed in Appendix D. For simulation-interface evaluation, RGB-based baselines follow their natural decode–encode pipeline: generated RGB frames are re-encoded by the frozen VLA tokenizer before token-space evaluation.

5.3 Evaluating Dynamics Modeling

We first evaluate the *dynamics-modeling axis*: given the same VLA visual-token prediction target, how should the transition dynamics be modeled? We compare Token-World with a deterministic Transformer-based token predictor adapted from DINO-WM [56]. Both methods predict future VLA visual tokens directly; the difference lies in the dynamics modeling choice.

Results. Table 1 and Fig. 4b show that Token-World improves VLA visual-token prediction across all tasks. The gain is most pronounced on long-horizon tasks: on *Stack-Bowls-Two*, RMSE drops from 1.8556 to 0.3712 and cosine similarity rises from 0.4531 to 0.9911 over the deterministic Transformer baseline. Ablations further validate the design choices: removing the GRU degrades rollout fidelity, while replacing x_0-prediction with v-prediction causes a larger drop, with RMSE increasing to 0.5818–0.6507 and cosine similarity falling to around 0.97. These results indicate that Token-World benefits not only from predicting VLA tokens directly, but also from more suitable dynamics modeling in this high-dimensional token space.

5.4 Evaluating the Simulation Interface

We next evaluate the *simulation-interface axis* by comparing Token-World with recent RGB-based world-model simulators [17, 38]. These baselines expose imagined futures as RGB observations and require VLA re-encoding for token-space evaluation, whereas Token-World directly rolls out future states in the VLA visual-token space. Ctrl-World [17] is included only for efficiency because it uses a large pretrained video diffusion backbone, while RMSE/cosine comparisons focus on from-scratch baselines.

Table 1: **Token fidelity under action-replay rollout.** We evaluate two complementary axes: dynamics modeling and simulation interface. For all methods, results are reported in the VLA visual-token space using trajectory-averaged RMSE (lower is better) and cosine similarity (higher is better).

Method	Grab-Roller		Place-Container-Plate		Place-Burger-Fries		Stack-Bowls-Two	
	RMSE↓	Cos↑	RMSE↓	Cos↑	RMSE↓	Cos↑	RMSE↓	Cos↑
Dynamics modeling								
DINO-WM [56]	0.4259	0.9874	0.6670	0.9807	1.7860	0.9741	1.8556	0.4531
Token-World w/o x_0-prediction	0.5818	0.9728	0.5801	0.9755	0.6295	0.9707	0.6507	0.9681
Token-World w/o GRU	0.2676	0.9957	0.3678	0.9910	0.3590	0.9919	0.4069	0.9887
Simulation interface								
WorldGym [38]	0.3947	0.9877	0.3485	0.9910	0.3606	0.9896	0.4102	0.9803
Token-World	**0.2620**	**0.9958**	**0.3236**	**0.9936**	**0.2790**	**0.9960**	**0.3712**	**0.9911**

Table 2: **Rollout-based success prediction under action-replay rollout.** We evaluate two complementary axes: dynamics modeling and simulation interface. We report Acc. (A), Prec. (P), Recall (R), and F1 for success/failure prediction on validation rollouts. Higher is better.

Method	**Grab-Roller**				**Place-Container-Plate**				**Place-Burger-Fries**				**Stack-Bowls-Two**			
	A	P	R	F1	A	P	R	F1	A	P	R	F1	A	P	R	F1
Dynamics modeling																
w/o x_0-prediction	0.67	**1.00**	0.25	0.40	0.50	0	0	0	0.45	0	0	0	0.50	0	0	0
w/o GRU	0.61	0.57	0.50	0.53	0.50	0	0	0	0.50	0	0	0	0.50	0	0	0
Simulation interface																
WorldGym [38]	0.33	0.17	0.13	0.14	0.50	0	0	0	0.50	0	0	0	0.50	0	0	0
Token-World	**0.94**	0.89	**1.00**	**0.94**	**0.60**	**1.00**	**0.20**	**0.33**	**0.70**	**1.00**	**0.40**	**0.57**	**0.65**	**0.71**	**0.50**	**0.59**

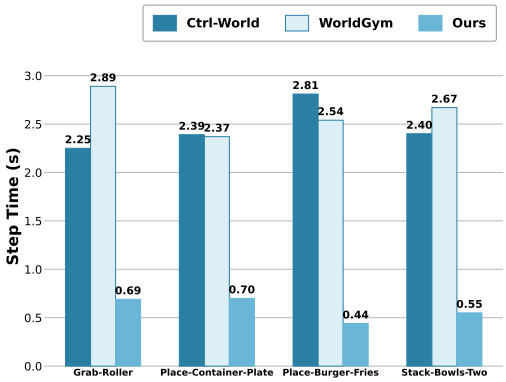
Results. Table 1 and Figure 4 shows that direct token-space simulation improves both rollout fidelity and efficiency. In token-space evaluation, Token-World achieves lower long-horizon RMSE than WorldGym [38] across all four tasks, e.g., reducing RMSE from 0.3947 to 0.2620 on *Grab-Roller*, from 0.3606 to 0.2790 on *Place-Burger-Fries*, and from 0.4102 to 0.3712 on *Stack-Bowls-Two*, while also improving cosine similarity. The efficiency gain is more pronounced: Token-World requires only 0.44–0.70s per simulator step, compared with 2.25–2.81s for Ctrl-World and 2.37–2.89s for WorldGym. This shows that avoiding RGB generation and VLA re-encoding substantially reduces the cost of every imagined interaction.

Table 2 further shows that Token-World improves overall success-prediction accuracy from 0.46 to 0.72 and F1 from 0.05 to 0.63 compared with WorldGym. On *Grab-Roller*, accuracy increases from 0.33 to 0.94 and F1 from 0.14 to 0.94. These results indicate that direct token-space rollouts preserve more task-relevant information than RGB-based decode–encode simulation.

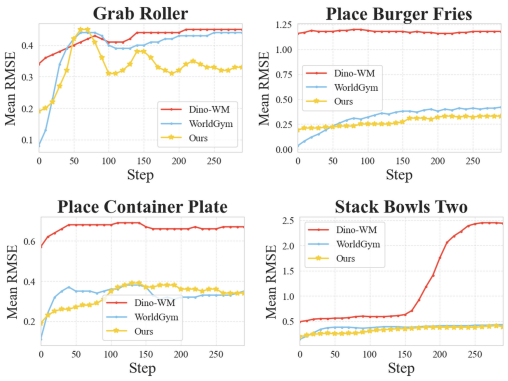
5.5 Qualitative Analysis

Fig. 5 visualizes patch-wise semantic changes in the Paligemma visual-token space. We compute the L2 distance from each timestep to the initial frame on a 16×16 token grid and upsample the map for visualization. The top row shows Token-World predictions, the middle row shows ground-truth token changes, and the bottom row shows RGB frames.

Token-World produces change maps that broadly match the ground-truth spatial patterns, with high responses concentrated around the robot arms and manipulated objects while background regions remain stable. This suggests that Token-World captures semantically meaningful action-conditioned changes rather than accumulating global token drift.



(a) Simulator step time comparison for the simulation-interface evaluation. Token-World directly predicts VLA visual tokens, avoiding RGB generation and re-encoding.



(b) RMSE over rollout steps across four RoboTwin [10] tasks. Token-World is compared with WorldGym [38] and DINO-WM [56] under action-replay rollout.

Figure 4: **Efficiency and long-horizon rollout fidelity.** Left: Token-World achieves lower simulator step time than RGB-based world-model baselines. Right: Token-World maintains lower RMSE growth over long-horizon rollouts compared with WorldGym [38] and DINO-WM [56].

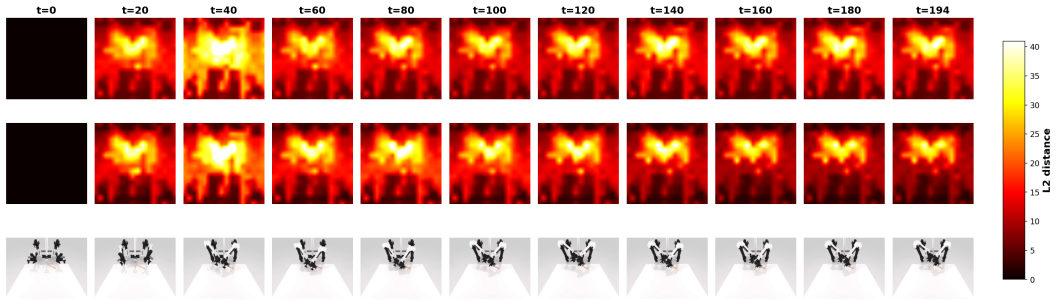


Figure 5: **Visualization of semantic token change.** We visualize per-patch semantic change from the initial frame, measured by the L2 distance in the Paligemma visual-embedding space and upsampled from a 16×16 patch grid. Top: predicted token rollout. Middle: ground-truth token trajectory. Bottom: corresponding RGB frames. Brighter regions indicate larger semantic deviation from the initial visual state.

6 Conclusion

We introduced Token-World, a VLA-oriented world-model simulator that rolls out future states directly in the policy-facing visual-token space. Unlike RGB-based simulators that reconstruct future images and then re-encode them into VLA inputs, Token-World avoids the decode–encode interface and directly models the representation consumed by downstream policies. To this end, Token-World combines a spatio-temporal multimodal Transformer, an x_0-parameterized flow-matching objective, and a recurrent state to capture high-dimensional token dynamics and preserve long-horizon interaction context.

Experiments on RoboTwin manipulation tasks validate Token-World along both dynamics-modeling and simulation-interface axes. Compared with deterministic token predictors and RGB-based world-model simulators, Token-World achieves better token prediction fidelity, lower simulator step time, and more reliable rollout-based success classification. These results suggest that aligning the simulation space with the downstream VLA policy is an important design principle for efficient and scalable world modeling. Future work will extend direct token-space simulation from action-replay evaluation to closed-loop policy evaluation, synthetic data generation, and policy learning for VLA agents.

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A Task and Dataset Details

We conduct all experiments on four simulated manipulation tasks from the RoboTwin [10] benchmark, instantiated on the Aloha-AgileX platform, a 14-DoF dual-arm robot in simulation. These tasks cover diverse bimanual coordination patterns and contact dynamics. The official RoboTwin task descriptions are as follows:

- **Grab-Roller**: Use both arms to grab the roller on the table.
- **Place-Container-Plate**: Place the container onto the plate.
- **Place-Burger-Fries**: Use both arms to pick the hamburger and french fries and place them onto the tray.
- **Stack-Bowls-Two**: Stack the two bowls on top of each other.

At each time step, the control input is represented as a 14-dimensional action vector obtained by concatenating the commands of the two arms, including both end-effector motion and gripper control. The proprioceptive state is represented using absolute end-effector states under the same per-arm parameterization, also resulting in a 14-dimensional vector. This shared representation enables a consistent interface between action and proprioceptive inputs during model training and rollout.

Table 3 summarizes the number of successful and failed episodes and their average trajectory lengths. Moreover, the train/validation split for successful and failed trajectories is reported in Table 4. During preprocessing, we compute the mean and standard deviation of the action and proprioceptive channels on the training split only, and use these statistics to normalize both training and validation data.

B Compute Resources, Hyperparameters and Implementation Details

In this section, we summarize the compute resources, hyperparameters and implementation details used in our experiments. Unless otherwise specified, all tasks share the same model architecture and optimization setup.

Table 3: Per-task dataset statistics.

Task	# Succ.	Succ. Len.	# Fail	Fail Len.
Grab-Roller	38	93.95	50	399.00
Place-Container-Plate	50	241.86	50	499.00
Place-Burger-Fries	50	158.54	50	399.00
Stack-Bowls-Two	50	313.54	50	899.00

Table 4: Train/validation split for successful and failed episodes.

Task	Train		Validation	
	Succ.	Fail	Succ.	Fail
Grab-Roller	30	40	8	10
Place-Container-Plate	40	40	10	10
Place-Burger-Fries	40	40	10	10
Stack-Bowls-Two	40	40	10	10

B.1 Compute Resources and Hyperparameters

All Token-World models are trained separately for each RoboTwin task. Unless otherwise specified, the model architecture uses hidden dimension 2048, 8 Transformer layers, 16 attention heads, head dimension 256, and 64 maximum denoising steps. Training uses batch size 8, Adam optimizer, learning rate 3.0×10^{-5}, and 100 epochs.

For each task, Token-World is trained with bfloat16 mixed precision for approximately 12 wall-clock hours on 8 NVIDIA A100 GPUs, corresponding to roughly 96 GPU-hours. Baseline models are trained under matched data and training-budget settings whenever applicable. Inference step time is measured on the same hardware for all methods. For RGB-based baselines, the reported step time includes both RGB generation and re-encoding with the frozen VLA visual tokenizer; for Token-World, it measures direct VLA visual-token generation. All timing results are averaged over validation rollouts. The architecture and training hyperparameters are summarized in Table 5.

B.2 Implementation Details

Since adjacent frames often differ only slightly, directly modeling every frame transition introduces substantial temporal redundancy. Following DINO-WM [56], we therefore introduce a frameskip parameter k. Concretely, given the observation at time step t, the model conditions on the sequence of k intermediate actions and directly predicts the observation at time step $t + k$. We use the *same* frameskip setting for DINO-WM and our method on each task for a fair comparison. We use a frameskip of 2 for the two shorter-horizon tasks, *Grab-Roller* and *Place-Container-Plate*, and a frameskip of 3 for the longer-horizon tasks, *Place-Burger-Fries* and *Stack-Bowls-Two*.

For action tokenization, we further split the 14-dimensional control vector by actuator type. Dimensions corresponding to delta end-effector motion are treated as continuous and encoded as a weighted sum of type-specific embeddings, i.e., $\sum_i \text{embed}_i \cdot \text{action}_i$, while the gripper dimensions are treated as discrete and encoded with learned embedding lookups. Thus, the action space remains a unified 14-dimensional control vector, but different subcomponents are tokenized differently according to their semantics. Proprioception is represented separately as a proprioceptive token. The predictor is implemented as an axial time-space Transformer.

C Decoder Architecture and Visual Metrics

C.1 Decoder Architecture and Training Details

To reconstruct pixel-space observations from visual tokens, we employ a **convolutional decoder with residual blocks and bottleneck self-attention**. Starting from low-resolution token features, the decoder progressively recovers spatial structure and appearance through residual convolution and hierarchical upsampling.

Table 5: Architecture and training hyperparameters. All tasks share the same setup except for task-dependent frameskip.

Model architecture

Parameter	Value
Hidden dimension	2048
Depth	8
Attention heads	16
Head dimension	256
Max denoise steps	64

Training hyperparameters

Parameter	Value
Batch size	8
Epochs	100
Learning rate	3.0×10^{-5}
Optimizer	Adam

Table 6: Decoder architecture and training hyperparameters.

Architecture hyperparameters

Parameter	Value
Input embedding dimension	2048
Base channel width	512
Channel multipliers	[1, 1, 2, 2, 4]
ResNet blocks per level	2
Output resolution	224×224
GroupNorm groups	32
Middle-block attention	True
Attention resolution	16
Output channels	3

Training hyperparameters

Parameter	Value
Batch size	8
Learning rate	3×10^{-4}
Learning rate schedule	Cosine annealing
Minimum learning rate	1×10^{-6}
Number of epochs	25
Latent loss weight	0.25

At the bottleneck resolution, the decoder applies an input convolution followed by two residual convolutional blocks with a self-attention block inserted between them. Each residual block contains two *GroupNorm* $\rightarrow$ *SiLU* $\rightarrow 3 \times 3$ *convolution* layers together with a skip connection; when the input and output channel dimensions differ, the skip path uses a 1×1 convolutional projection for dimension matching. The self-attention block is applied only at the bottleneck resolution to model global spatial dependencies before decoding to higher resolutions.

The bottleneck features are then passed through four decoding stages. Each stage contains two residual blocks followed by a $2\times$ spatial upsampling operation implemented by nearest-neighbor interpolation and a 3×3 convolution. Across these stages, the spatial resolution is progressively increased, while the channel dimension is reduced from 2048 to 512.

The output head uses GroupNorm, SiLU, and a final 3×3 convolution to produce a 3-channel RGB image, followed by a `tanh` activation. The architecture hyperparameters and training hyperparameters are summarized in Table 6.

C.2 Qualitative Results and Visual Metrics

We further evaluate the decoder both qualitatively and quantitatively. Figure 6 shows representative reconstruction results on validation trajectories. For each example, the first row presents the ground-truth frames, and the second row shows the images reconstructed from latent visual tokens.

For quantitative evaluation, we report standard visual reconstruction metrics in Table 7. In addition to distortion-based metrics such as PSNR and SSIM, we also report the perceptual metric LPIPS to better reflect visual fidelity.

D Latent Success Classifier Details

In this section, we provide the implementation and training details for the latent success classifier. Crucially, aligning with the core motivation of Token-World, our success classifier operates *directly* on the VLA visual token representations rather than raw RGB pixels. This design avoids the computational overhead of decoding imagined latents back to images and yields a highly efficient evaluator.

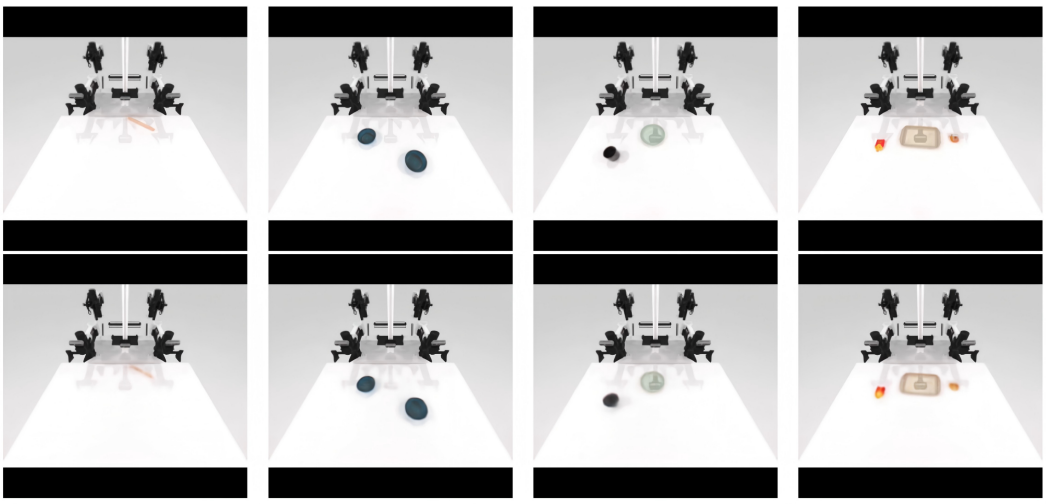

Figure 6: **Qualitative reconstruction results of the decoder.** Representative examples from the four RoboTwin tasks. The first row shows the ground-truth frames, and the second row shows the corresponding images reconstructed from latent visual tokens.

Table 7: **Visual reconstruction metrics of the decoder on the validation set.** All metrics are computed over the full validation set, containing 29,850 frames in total. Higher PSNR/SSIM and lower LPIPS are better.

Task	PSNR↑	SSIM↑	LPIPS↓
Grab-Roller	35.13	0.9874	0.0121
Place-Container-Plate	36.02	0.9880	0.0115
Place-Burger-Fries	36.39	0.9893	0.0125
Stack-Bowls-Two	36.10	0.9878	0.0102
Average	35.91	0.9881	0.0116

D.1 Architecture and Data Processing

The latent success classifier is a lightweight multi-layer perceptron (MLP) with approximately 1.1M parameters. It takes the pre-extracted VLA token latents of shape 256×2048 (number of patches $\times$ embedding dimension) as input. The network first applies spatial mean pooling across the patch dimension to aggregate the features into a single 2048-dimensional global representation. This is followed by a 3-layer MLP ($2048 \to 512 \to 128 \to 1$) with Layer Normalization and ReLU activations. A final Sigmoid activation outputs the success probability $p \in [0, 1]$. The entire model contains only $\sim$1.1M trainable parameters, making it extremely fast for both training and inference.

To train the classifier, we utilize the same four manipulation tasks from the RoboTwin benchmark. We split the episodes into 85% for training and 15% for validation at the trajectory level to prevent data leakage. For successful trajectories, we label the last 20% of frames as positive samples (success, $y = 1$) and the first 80% as negative samples (in-progress, $y = 0$). For failure trajectories, all frames are labeled as negative. To address class imbalance, we dynamically sample frames during training and apply a $3.0\times$ positive sample weight in the loss function.

D.2 Training Details

The model is trained using Binary Cross-Entropy (BCE) loss. We use the AdamW optimizer with a cosine annealing learning rate schedule following a brief linear warmup. Detailed hyperparameters are summarized in Table 8.

Table 8: **Latent success classifier training hyperparameters.** Configuration for training the success classifier directly on VLA visual tokens.

Hyperparameter	Value
Input shape	256×2048 VLA token latents
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Batch size	64
Training epochs	30
Learning rate schedule	Cosine annealing with 2-epoch linear warmup
Positive sample weight	3.0

D.3 Validation Performance

Despite having approximately $10\times$ fewer parameters than a standard ResNet-18 pixel-based classifier, the latent success classifier demonstrates strong discriminative capability. It achieves its best validation performance at epoch 28. As shown in Table 9, the model balances precision and recall effectively, yielding an F1 score of 90.67% and a high recall of 93.38%, ensuring reliable and fast success-detection signals during Token-World policy evaluation.

Table 9: **Latent success classifier validation results.** Binary classification metrics evaluated on the held-out validation set with 5,194 latent samples.

Accuracy	Precision	Recall	F1 Score
95.48%	88.12%	93.38%	90.67%

E Assets and Licenses

We use existing research assets only for simulation, tokenization, and baseline comparison. RoboTwin is used as the simulated manipulation benchmark and is released under the MIT license. The VLA visual tokenizer is based on PaliGemma/SigLIP components: PaliGemma access requires agreeing to Google's usage license, and the public SigLIP model card lists an Apache-2.0 license. We also use or compare against existing research codebases and models, including π_0/OpenPI, DINO-WM, WorldGym, and Ctrl-World, following their intended research use. We cite all external assets in the main paper and do not redistribute their datasets, model weights, or simulator assets as part of this submission.

F Broader Impacts and Limitations

Token-World studies how learned world models can simulate future robot states more efficiently in the representation space consumed by downstream VLA policies. A potential positive impact is to reduce the cost of embodied learning: if imagined rollouts can be generated directly in a policy-facing representation space, they may support faster policy evaluation, data filtering, and future policy improvement with fewer physical robot trials. This can be especially useful in robotics, where real-world data collection is expensive, slow, and may cause hardware wear.

At the same time, learned simulators can introduce risks when their predictions are treated as reliable substitutes for real interaction. A world model may produce plausible but incorrect rollouts, and such errors may bias downstream policy evaluation or data generation. This risk is particularly important for long-horizon manipulation, where small prediction errors can compound over time. In this work, we therefore evaluate Token-World under action-replay rollouts and rollout-based success prediction, rather than claiming full closed-loop VLA policy improvement.

Token-World is currently evaluated in simulation on RoboTwin manipulation tasks. The conclusions should not be interpreted as evidence that direct token-space simulation is sufficient for safe real-world

robot deployment. Before being used for policy optimization or real-robot decision making, such simulators should be validated under closed-loop settings, tested across broader task distributions, and combined with uncertainty estimation or real-world verification. We view this work as a step toward more efficient VLA-oriented world modeling, rather than a complete replacement for physical evaluation.

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